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Automatic License Plate Recognition System Using SSD

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***Abstract*—Automatic License Plate Recognition (ALPR) is a very widely used system in applications such as parking management, theft detection, traffic control and management etc. Most of the existing ALPR systems fail to showcase acceptable performance on real time images/video scenes. This work proposes and demonstrates implementation of a deep learning-based approach to locate license plates of four wheeler vehicles thereby enabling optical character recognition (OCR) to recognize the characters and numbers on the located plates in real time. The proposed system is decomposed into three sub-blocks viz. Vehicle image/video acquisition, License plate localization and OCR. A simple setup using a reasonable resolution webcam has been designed to capture images/videos of vehicles at some entry point. We propose to utilize Single Shot Detector (SSD) based Mobilenet V1 architecture to localize the license plates. The hyper parameters of this architecture are selected with rigorous experimentation so as to avoid over-fitting. We have compared performance of two OCRs viz. Tesseract OCR, Easy OCR and found the superiority of Easy OCR since it utilizes deep learning approach for character recognition. NVIDIA Jetson Nano and Raspberry Pi 3B hardware platforms have been used to implement the entire system. The parameters of these three sub-blocks have been optimized to yield real time performance of ALPR with acceptable accuracy. The proposed and implemented system on Jetson Nano allows processing of videos for ALPR having accuracy more than 95%.**

***Index Terms*—SSD, EasyOCR, Tesseract OCR**

I. Introduction

An automatic license plate detection and recognition system is the combination of hardware plus software to identify the license plate number of a particular vehicle. Initially, the proposed method identifies the license plate of the vehicle. The essential preprocessing is implemented on the identified picture before passing it to the recognition phase. The Optical Character Recognition (OCR) is employed in the recognition phase. The OCR output is saved in the database for monitoring. Numerous circumstances that can influence the performance of the system are the speed of a vehicle, angle of capture, language of the license plate number, different environmental conditions, and number plates with unnecessary creativity. The significant application of this system is in parking monitoring, Automatic toll collection, Theft detection, Traffic monitoring. The proposed framework has two important stages viz. Detection/Localization and Recognition. The detector decides the occupancy of the License Plate area in the frame captured by the camera module. The recognizer reads the text from the located license plate. Since detection accuracy plays a major role in recognition for the ALPR system, our paper offers an efficient system with the foundation of Single Shot detector architecture. The proposed and implemented system is based on the Indian License Plates. Our framework uses EasyOCR and Tesseract OCR as a recognizer. The whole system is implemented and tested on NVIDIA's Jetson Nano and Raspberry Pi 3B.

II. Related Work

The implementation of the ALPR system is based on two stages viz. Detection and Recognition. The conventional way of detecting license plates majorly depends on edges, texture, shape, etc. In [1] the advanced Vertical edge detection algorithm(VEDA) was proposed. This algorithm is based on the contrast between grayscale values and it is efficient for ALPR detection. Authors of [2] have used vertical edge detection algorithm for the detection of the license plates and for recognition the cross-correlation based template Matching mechanism is utilized. In template matching the obtained output and pre-stored data is correlated to give the final results. In [3] the authors have utilized the Morphological operators in the process of detection. These operators are sensitive to the shape, Therefore the rectangular operator was employed. In recognition, template matching is used. In [4] to improve the detection rate the cascade classifiers were implemented to get the efficient detection result and for recognition, authors have utilized the Tesseract OCR library. Apart from these conventional approaches, the techniques involving CNN are also utilized in the ALPR detection. authors of [5] employed Faster R-CNN in detection to achieve better results. The method involved in [6] is designed for Chinese License plates. Therefore, the improved LeNet-5 is implemented for recognition. The improvement is done to match the requirement of the Chinese language. The Yolo v2 is trained for the detection in [7] whereas for recognition combination of ResNet-5 and Mask R-CNN is utilized.

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The License Plate localization in [8] is implemented by combining two CNN models. The deep learning(SSD) based CNN system is proposed and implemented in this paper to reduce the detection time and hence can be effectively utilized in the real-time implementation of the system.

III. THE PROPOSED METHOD

The generic architectural flow of the ALPR system is shown in figure 1.Detection and recognition are two pillars of the ALPR system. To make the system more precise, deep learning and optical character recognition methods are employed.

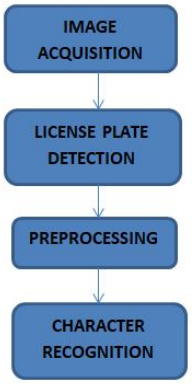

Fig. 1. System Framework.

A. License Plate Detection

Image acquisition is the primary step in the License Plate Detection procedure.Image acquisition is the method of capturing the input for the ALPR system. In the proposed methodology, we have utilized the webcam to fulfill the need for endless input. The webcam i.e. camera module captures the frames continuously.To obtain reasonable image features the resolution selected is 1280x720. The camera module used here is ViewCam Tango. The captured frames are supplied to the detector module without any preprocessing.Conventional methods for detection mainly involve detection/classification based on the color, texture, and edges.The accuracy in a conventional way for real-time implementation is not best. To improve the accuracy of the model, the best approach is to employ deep learning models.

1) Single Shot Detector: The framework depends on Single Shot Detector architecture.Deep learning-based SSD architecture is faster compared to the different deep learning models.SSD doesn't have a proposal for a bounding box [9]. This results in an improvement in speed. The core framework of a single-shot detector is employed to predict the classification scores and offsets of boxes. To acquire the best returns convolutional filters/kernels of small dimensions are applied to the feature map [9]. At every layer of feature convolutional model utilized is changed/different [9]. Due to this highly accurate model with reasonable and acceptable latency is achieved. SSD-based MobileNet version 1 is utilized as a detector. MobileNet is greatly lightweight and simpler to execute on hardware. MobileNet is used in real-time applications because of its lightweight nature. MobileNet is based on the structure that uses Depthwise Separable Convolution to build a lightweight CNN [10]. MobileNet is effective network architecture. It has two hyperparameters that produce very small and low latency models [10]. MobileNet model characteristics can be utilized to implement mobile and embedded applications [10]. To achieve the commending results specific parameters need to be optimized. After rigorous experimentation, the parameters were optimized.

The parameters to be optimized are:

1) Number of classes:
   The number of classes symbolizes the number of foreground classes that the classifier needs to classify. In the proposed system License Plate is a region of interest. Everything which is not a license plate will be considered as a background class. Hence, the number of classes is set to 1.

2) Learning Rate:
   The learning rate is a configurable hyperparameter. It is utilized in the training of CNN. Usually, the learning rate has a positive value in a set of (0,1). The learning rate manages how fast the model is adjusted to the problem. The decaying exponential learning rate updates its value until the local minimum is achieved. The high converge speed is achieved with the use of decaying exponential. Hence, The decaying exponential learning rate is utilized.

3) Width Multiplier [10]:
   It is also recognized as the depth multiplier. Depth multiplier makes the network thin evenly at each layer. By modifying this parameter network size can be reduced. Due to the depth multiplier, the smaller model with reasonable accuracy and latency tradeoff is obtained. Its default value is 1. The decrease in the value of the width multiplier reduces the network size, but the accuracy decreases [10]. Hence default value is used in this work. It is denoted by alpha.

4) Resolution Multiplier [10]:
   The resolution multiplier parameter decreases the computational cost of the model. Value can be optimized in range (0,1]. Due to the resolution and width multiplier, MobileNet is selected. To shorten the model, resolution and width multiplier are used. The default value of the resolution multiplier is 1. The decrease in the value of the resolution multiplier reduces the network size, but the accuracy decreases [10]. Hence default value is used in this work.

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5) Train-Test Split:
The 80-20 split method is selected for testing the accuracy. 640 images out of 800 were selected as training set and 160 images were kept in the testing dataset. Train and test records for the same are generated and used in process of training.

The layered structure is obtained after training the model with the optimized and precise parameters. The layered structure is shown in figure 2.

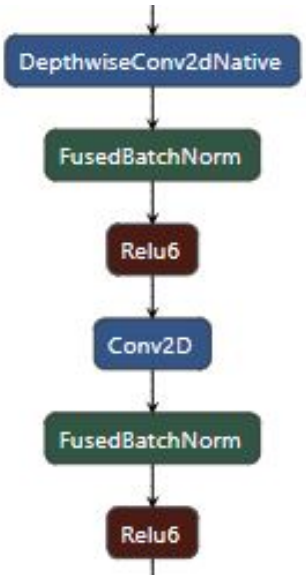

Fig. 2. Layer-wise Network Architecture. [10]

MobileNet is developed on depthwise separable convolution. The depthwise separable convolution operates in two steps viz. depthwise convolution and pointwise convolution [10]. All the layers of MobileNet SSD are followed by Batch Normalization and RELU nonlinearity [10]. Softmax Layer is employed for classification. The MobileNet model has almost all of the computation in the pointwise convolution layer [10]. The image transformation does not happen in pointwise convolution layer. Hence, Most of parameters are in pointwise layer.

The output of the detector is the coordinates of the License Plate area. These coordinates are supplied as input to the preprocessing section.

B. Preprocessing

The coordinates received from the license plate detector are utilized for essential preprocessing. The essential preprocessing involves the cropping of the detected license plate area. The cropped image in this framework undergoes image enhancement and image blur filter procedures to enhance the accuracy of OCR. Figure 3 displays the cropped image at the output of the preprocessing.

Fig. 3. System Framework.

The cropped image in figure 3 has cheap resolutions and it can influence the accuracy of the recognizer. To avoid this shortcoming essential preprocessing step is included in the proposed framework.Image scaling, image enhancement are implemented on the cropped image.The output achieved following essential preprocessing is displayed in figure 4. The output of the preprocessing stage is served to the recognizer to give a final output of the system.

Fig. 4. Cropped image after scaling and enhancement.

C. Recognition

The recognition is the final pillar of the ALPR system. Optical Character Recognition (OCR) is employed in the process of recognition. The task of OCR is to read letters and numbers from the enhanced license plate obtained from the Preprocessing stage. In this proposed and implemented system, two OCR methods are implemented and comparative analysis is obtained.

1) Tesseract OCR: In Tesseract OCR processing of cropped frame obeys systematic pipeline. At the primary level components of the outline are obtained and stored using connected component analysis. Outlines are nested together to make the blobs. Blobs are arranged into text lines. After fixed and proportional region analysis the text line is separated into words according to character and region spacing. There is an adaptive classifier at the end for accurate recognition in Tesseract OCR. [11]

2) Easy OCR: It is an OCR method supporting 80+ languages. It recognizes characters from the image. The easy OCR is based on the ResNet, LSTM, CTC models for the recognition of characters. The detection part in the EasyOCR is done using Craft Algorithm. EasyOCR has three main components. The first is feature extraction. The ResNet model is trained for feature extraction. The second is Sequence labeling. For sequence labeling LSTM (Long Short-Term Memory) algorithm is used and the final component is decoding. Decoding is based on CTC (Connectionist Temporal Classification). Readtext function of EasyOCR is utilized in the process of recognition. The main feature of easy OCR is that it reads letters and numbers from images and returns the coordinates where it was located. [12]

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We have proposed the approach involving Tesseract OCR and Easy OCR as recognition method. Both of the methods are implemented separately.

IV. EXPERIMENTATION & RESULTS

The proposed system is implemented with the purpose of functioning in a real-time environment. To achieve the best results several experiments are performed on the system. While implementing the system we found several factors which were affecting the performance of the system. The problems that occurred, their solution, and results are addressed in this section. Our proposed system is implemented on the basis of Single-shot detector architecture. The whole system is implemented in Python language. For software implementation of the ALPR system, the Anaconda Spider tool is utilized. The developed framework is implemented and tested on two hardware platforms. Raspberry Pi 3B and NVIDIA's Jetson Nano is employed as a hardware device. The comparative analysis is explained in this section.

A. Detection

The optimized parameters to get the best results are discussed in section 3. Initially, the trained model is implemented for the image input. The results achieved are shown in Figure 5. After examining the model on image input it

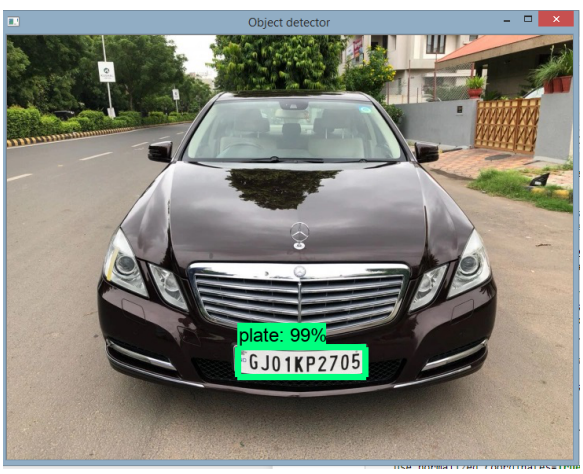

Fig. 5. Detection Results for image input.

has experimented for real-time input.Figure 6 and Figure 7 shows the output for real-time data.

In Figures 6 and Figure 7, it can be observed that the camera is directly faced at the vehicle. To examine the efficiency of the model it has experimented on the real-time input with a slanted camera angle. The results achieved after fixing the camera at several angles are presented in Figure 8.

After experimenting with all scenarios that could affect the detector accuracy, the model was tested on the test

Fig. 6. Detection result for real-time input.

Fig. 7. Detection result for real-time input.

set images to calculate the accuracy. Table 1 shows the detection accuracy. In table 1:

Set 1 : This set contains 50 images captured at a straight angle without an inclination. The environmental condition for this set of pictures was normal.

TABLE I
DETECTION ACCURACY

Set No.	No. of images	Correct Detection
Set No. 1	50	49
Set No. 2	50	46
Set No. 3	56	53

Set 2 : This set includes 50 images. The images in this set are with various orientation. The plates in this set are from different states other than Maharashtra.

Set 3 : This set involves vehicle images captured from different angles and distances. The number of images in this

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Fig. 8. Detection result for slanted camera angle.

set is 56. The output of the detector model is coordinates of the LP region. Figure 9 presents the output of the detector.

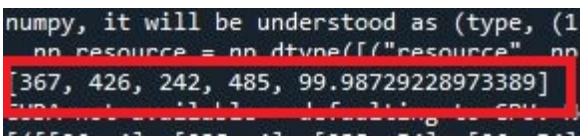

Fig. 9. Detector output.

B. Recognition

After experimentation of detection, the whole system is integrated with OCR to get terminal results. The recognition process is done by using two techniques i.e. Tesseract OCR and Easy OCR. The OCR result using Tesseract OCR is shown in figure 10. The result is achieved for the image in figure 5.

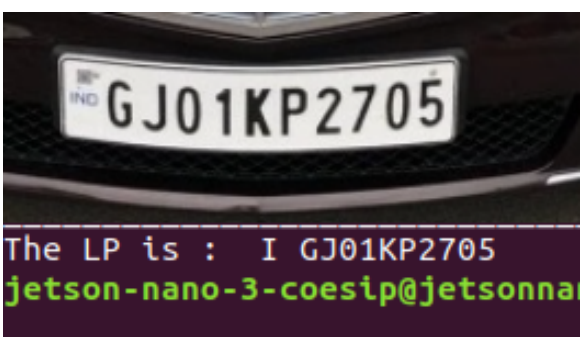

Fig. 10. Tesseract OCR output.

Accuracy results obtained for the Tesseract OCR are presented in table 2. After testing tesseract OCR, the results for the images using Easy OCR were examined. Figure 11 presents the output of Easy OCR. As shown in figure 11, Easy OCR reads the text as well as provides the coordinates of text where it was located. To test the correctness of Easy OCR it is executed on the set 78 images. Table 3 gives an accuracy of Easy OCR.

TABLE II
TESSERACT OCR ACCURACY

Set No.	No. of images	Correct Recognition
Set No. 1	26	18
Set No. 2	26	17
Set No. 3	26	14

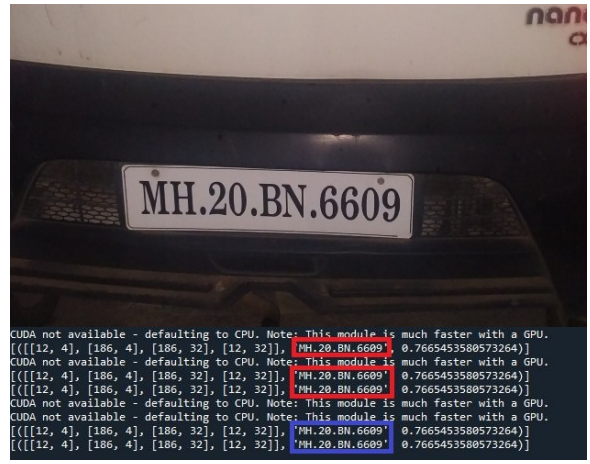

Fig. 11. Easy OCR output.

TABLE III
EASYOCR ACCURACY

Set No.	No. of images	Correct Recognition
Set No. 1	26	24
Set No. 2	26	22
Set No. 3	26	24

After experimenting with Tesseract and Easy OCR, Comparative analysis is discussed in table 4. Accuracy for both OCR methods is obtained after experimenting with a same set of test images.

TABLE IV
TESSERACT OCR V/S EASYOCR

Parameter	Tesseract OCR	Easy OCR
Returns	Text/String	Numpy Array
Accuracy	62%	90%
Recognition Time	0.6 sec	0.84 Sec
Raspberry Pi	Supported	Not Supported
Jetson Nano	Supported	Supported

C. Hardware Platform

After rigorous experimentation on raspberry pi 3B to get better performance, better fps and to avail benefits of GPU, the model is deployed on the Jetson Nano platform. The comparative analysis of both hardware platform is presented

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in table 5. Figure 12 shows the whole system output on Jetson Nano. Jetson nano supports both of the OCR techniques. Whereas Raspberry Pi 3B supports only Tesseract OCR. EasyOCR needs PyTorch version >=0.5 which is not supported by Raspberry pi 3B hence we were not able to implement EasyOCR on Raspberry Pi 3B.

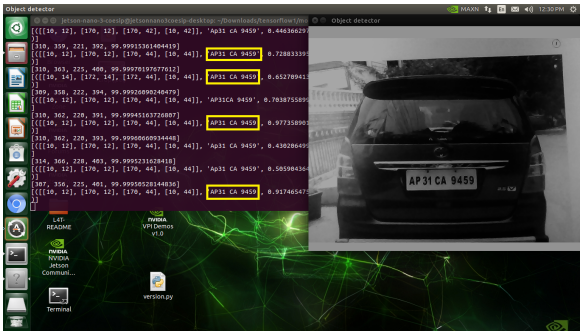

Fig. 12. Jetson Nano Implementation.

TABLE V
JETSON NANO V/S RASPBERRY PI

Parameter	Jetson Nano	Raspberry Pi
OS	linux based OS	Raspbian OS
RAM	4 GB	1 GB
Detection Time	0.09 Sec	4 Sec
Recognition Time	0.91 sec (Tesseract) 0.84 sec (Easy OCR)	0.8 sec (Tesseract)
Tesseract OCR	Supported	Supported
Easy OCR	Supported	Not Supported
Heating issues	More heating issues	Less Heating issues

V. CONCLUSION

In this paper, a method for automatic license plate detection and recognition is presented. It can be concluded that the SSD-based detector works efficiently with an accuracy of 94.87 %. The detector can detect the license plate with different a perspective. In this system, two efficient OCR techniques are implemented. From the experimental results, it can be inferred that the Easy OCR gives better accuracy than Tesseract OCR. The accuracy of Tesseract OCR is 62 % whereas Easy OCR operates with an accuracy of 90%. The results obtained are on the images captured in bright and overcast conditions. The proposed system gives significant results in bright and overcast environmental conditions compared to night conditions.The system is implemented and tested on two hardware platforms namely Jetson Nano and Raspberry Pi 3B. Due to the inclusion of AI and Computer vision libraries in Jetpack, the performance of Jetson Nano is better compared to the Rasberry pi 3B. The SSD-based detector along with Easy OCR implemented on Jetson Nano gives better accuracy and performance.

VI. CONTRIBUTION

We proposed and implemented a detector model based on the Single Shot Detector architecture. We proposed and implemented the concept of integration of Easy OCR with SSD. The real-time implementation of the system on Jetson Nano platform is achieved in this proposed system.

VII. FUTURE WORK

Future work can be performed by training a model for non-standard license plates. The recognition time and accuracy can be improved as an extended part of this work. Detection for very high-speed vehicles can also be examined. The accuracy of the proposed system can be quantified by confusion matrix, precision and recall values.

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